



The Relationship Between the Implementation of Environmental Protection Tax Law Policy and Enterprise Financial Performance—Empirical Analysis Based on Big Data Technology

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Abstract: In order to alleviate environmental problems, China has formulated the environmental protection tax law. However, how does its implementation affect the financial performance of enterprises? This paper selects A-share listed companies from 2012 to 2024 as the research sample from the massive data of China Stock Market & Accounting Research (CSMAR) database, and empirically studies the relationship between the implementation of environmental protection tax policies and corporate financial performance by using difference-in-differences big data analysis technology. The study found that the implementation of the environmental protection tax law policy is positively correlated with the financial performance of enterprises, and after a variety of data visualization robustness tests, the conclusion is still valid. Through the analysis of big data mechanism, we found that technological innovation and agency cost play a certain mediating effect between the two. In order to ensure the implementation of environmental protection tax, it is suggested to build a cross sectoral risk warning mechanism based on data sharing, and rely on the dynamic tax preference model to realize “the less pollution, the more preferential”, so as to stimulate the green innovation power of enterprises.

Keywords: Environmental protection tax; Policy implementation; Big data analysis technology; Empirical research; Analysis of big data mechanism

1. Introduction

China's economy is growing (Yu et al., 2024) and people's living standards are improving (Li & Hua, 2024), but economic growth has caused environmental pollution (Wang & Shao, 2019; Wu et al., 2016). The proportion of China's carbon dioxide emissions in the global total is rising (He et al., 2010). People have to pay attention to environmental problems (Ren et al., 2024). The Chinese government attaches great importance to environmental issues (Liu et al., 2024), and has issued a series of environmental laws and regulations (Jiang et al., 2023). In the 1970s, China set up a pollution charge, but the system has some defects (Wang & Zhang, 2024). After that, the environmental protection tax law of the People's Republic of China was issued in 2016 and officially implemented on January 1st, 2018 (Li et al., 2021). The introduction of environmental protection tax law is conducive to the harmonious development of enterprises and ecology (Li et al., 2024).

Enterprises, as the units that discharge pollution, are the key regulatory objects of environmental protection tax. The collection of environmental protection tax will affect the production activities of enterprises and thus affect their financial performance. Most enterprises take profit maximization as their business objective. Therefore, when the tax cost of environmental protection tax is lower than the cost of pollution remediation, enterprises will tend to maintain the original production mode and not treat pollution. When the cost of environmental protection tax increases, the income obtained by enterprises maintaining the original extensive generation method will not be

able to pay the environmental protection tax. At this time, the collection of environmental protection tax can promote enterprises to improve the production process, improve the pollution treatment level through various ways, and finally realize the coordinated development of enterprise benefits and environmental protection.

Therefore, this paper treats the implementation of the environmental protection tax policy as a quasi-natural experiment and employs a difference-in-differences approach to examine the relationship between the implementation of the environmental protection tax policy and enterprise financial performance. Furthermore, a mechanism analysis is conducted to investigate the mediating mechanisms underlying this relationship. The findings are expected to provide useful insights for improving China's fiscal and tax system and promoting the innovation and development of its environmental tax system.

2. Literature Review and Research Hypotheses

Although China's economy is growing rapidly, it is characterized by high investment and heavy pollution (Chen et al., 2025). This growth model makes China's environmental problems increasingly prominent (Zhou & Lin, 2025). Achieving environmental protection alongside economic development requires appropriate environmental regulations (Mei & Zhang, 2025). The Chinese government attaches great importance to environmental remediation and has formulated a variety of regulations and policies for environmental protection (Luo et al., 2025). As an important environmental regulation, environmental taxation has contributed a lot to environmental protection (Jiang et al., 2024). In order to better play the important role of environmental taxation, the environmental protection tax law was formally implemented in 2018 (Kong et al., 2024). Levying environmental tax will transfer the external environmental costs of enterprises to the internal. This measure has greatly stimulated the initiative of enterprises' green transformation, and also made the market incentive and regulation mechanism effectively used in the field of environmental protection (Duan & Rahbarimanesh, 2024).

Neoclassical economic theory believes that the collection of environmental protection tax will increase the tax cost of enterprises, which will have a negative impact on the financial performance of enterprises. However, Porter (1991) put forward a completely different view. He believed that when the environmental protection tax was appropriately stringent, enterprises would greatly improve their innovation initiative, and the innovation compensation income generated would be sufficient to make up for the environmental tax cost, which would ultimately promote the improvement of enterprise financial performance (Porter, 1991). The previous level of pollution charge was low, and many enterprises were unwilling to improve pollution control technology even if they paid the fee. The environmental protection tax law implemented in 2018 has improved the collection standard, which can essentially improve the effect of environmental regulation policy, making enterprises have to improve environmental protection production technology, so as to better play the incentive effect of environmental regulation (Freire-González, 2018). The enforcement of the environmental protection tax law has greatly reduced the rent-seeking behavior of enterprises, and the improvement of tax standards has also "forced" enterprises to improve their technical level and reduce environmental pollution. In order to protect the financial performance of enterprises from the adverse impact of tax costs, enterprises will improve pollution treatment technology in various ways, so as to reduce environmental tax costs. On the other hand, the collection of environmental protection tax can not only force enterprises with high pollution and low efficiency to exit the market, but also promote enterprises with high environmental and economic benefits to enter the market, so as to promote the overall improvement of enterprise financial performance. According to these theories, this paper puts forward Hypothesis 1.

Hypothesis 1: There is a positive correlation between the implementation of environmental protection tax law and corporate financial performance.

In terms of tax calculation standard, the environmental protection tax law takes the previous discharge fee collection standard as its lower limit, while the upper limit of tax collection has been improved to a certain extent. In order to give full play to their subjective initiative, each province can choose its own tax rate according to its own economic development. When the tax rate is appropriate and reasonable, the enterprise will improve the level of technological innovation, reduce environmental pollution and optimize the production process, and ultimately promote the improvement of enterprise financial performance. On the other hand, the environmental protection tax has also set up tax relief policies. The tax relief policy can encourage enterprises to improve the level of technological innovation to make their pollution emissions meet the requirements of tax relief, so as to reduce the tax cost and promote the rise of enterprise financial performance. Based on these analyses, this paper proposes Hypothesis 2.

Hypothesis 2: The implementation of environmental protection tax policy promotes enterprises to enhance the level of technological innovation and enhance their financial performance.

Due to the separation of ownership and management rights in modern enterprises, the interests of shareholders and operators are not exactly the same. Operators may erode environmental protection funds due to profit seeking, muddle along with environmental protection and governance, resulting in high agency costs for enterprises. The implementation of the environmental protection tax law is mandatory and universal at the legal level, which can

effectively prevent the rent-seeking measures of operators, reduce the agency costs caused by the information gap, and play the inhibitory role of the implementation of the environmental protection tax law policy on the agency costs of enterprise operators in pollution remediation, so as to promote the improvement of enterprise financial performance. Therefore, this paper proposes Hypothesis 3.

Hypothesis 3: The implementation of environmental protection tax policy can reduce the agency cost of enterprises and improve the financial performance of enterprises.

Drawing on previous literature on big data analysis and mechanism analysis (Giordino & Crocco, 2025; Hong et al., 2025; Sun et al., 2019; Zhang, 2019), the following empirical analysis employs a difference-in-differences approach to test Hypothesis 1 and a big data-based mechanism analysis approach to test Hypotheses 2 and 3.

3. Big Data Sample Screening and Research Design

3.1 Big Data Sample Screening and Data Sources

This paper selects China's A-share listed enterprises from 2012 to 2024 as the research object from the massive data of China Stock Market & Accounting Research (CSMAR) database, and uses big data management technology to divide the sample enterprises into the implementation group which is more deeply affected by the implementation of environmental protection tax policies and the control group which is less affected by it. Heavily polluting enterprises are deeply affected by the environmental protection tax law. According to the discussion on heavy pollution industries in the guidelines for *Industry Classification of Listed Companies* and the guidelines for *Environmental Information Disclosure of Listed Companies*: Heavy pollution industries refer to those industries that consume a lot of energy and discharge a lot of waste in the production process, cause significant pollution to the environment, and are clearly defined as key regulatory industries by the national environmental protection department. According to the classified management directory of *Listed Companies' Environmental Protection Verification Industry (HBH [2008] 373)*, thermal power, steel, cement, electrolytic aluminum, coal, metallurgy, chemical industry, petrochemical, building materials, papermaking, brewing, pharmaceutical, fermentation, textile, tanning and mining industries are defined as heavy pollution industries, according to the sample of this article, we select Coal mining and washing, oil and natural gas extraction, ferrous metal mining and beneficiation, non-ferrous metal mining and beneficiation, textile industry, leather, fur, feather and down products industry, paper and paper products industry, petroleum processing, coking and nuclear fuel processing industry, raw chemical materials and chemical products manufacturing, pharmaceutical manufacturing, chemical fiber manufacturing, rubber products industry, plastic products industry, non-metallic mineral products industry, ferrous metal smelting and rolling processing industry, non-ferrous metal smelting and rolling processing industry, and electric power and heat production and supply industry for heavy pollution industries. Listed companies in these 17 industries are the implementation group, while other enterprises are the control group. In order to test whether the grouping in this paper is reasonable, we conducted a parallel trend test. We applied data visualization technology to list the results of the parallel trend test in Figure 1.

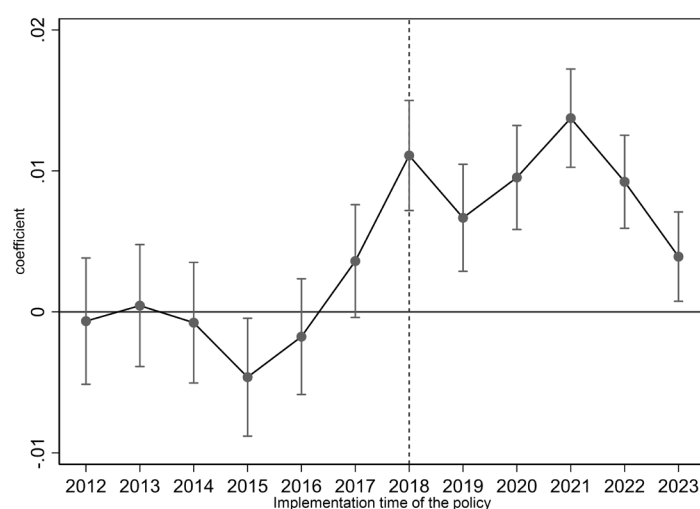


Figure 1. The results of parallel trend test

The environmental protection tax law was formally implemented in 2018. In Figure 1 presented by data visualization technology, the confidence intervals for the regression coefficients of the enterprise financial

performance variables from 2012 to 2017 almost include zero, that is, before the implementation of the environmental protection tax law, there was no significant difference in the enterprise financial performance between the implementation group and the control group. In 2019–2023, the confidence interval of the variable coefficient of enterprise financial performance will no longer include 0, that is, after the implementation of the environmental protection tax policy, the enterprise financial performance of the implementation group and the control group has significant difference, which proves that our grouping is relatively reasonable.

In this paper, the samples were further screened by using big data screening technology with reference to previous literature: (1) Deleted financial enterprises; (2) Delete the enterprise with special treatment; (3) The obvious abnormal data are deleted. For example, the shareholding ratio of the largest shareholder exceeds 100%; (4) 1% and 99% tail reduction processing were performed on all continuous data. Finally, we obtained 55066 observations. The data source is CSMAR database, and the empirical analysis tool is big data analysis software STATA 17.0.

3.2 Big Data Variable Design

3.2.1 Dependent variable

The dependent variable of this paper is enterprise financial performance. Referring to the previous literature, this paper selects the total net asset interest rate (*ROA*) as the dependent variable, and applies the return on equity (*ROE*) as the financial performance index of the robustness test for the robustness analysis.

3.2.2 Independent variable

The independent variable of this paper is the implementation effect of the environmental protection tax policy. According to the difference-in-differences analysis, it is expressed as $Dif = Im \times a_b$, where *Dif* represents the interaction term between *Im* and *a_b* and captures the effect of the environmental protection tax policy. *a_b* indicates the period before and after the implementation of the environmental protection tax policy; that is, the value of *a_b* is 1 after 2018 and 0 before 2018. *Im* indicates whether an enterprise belongs to the treatment group. The value of *Im* is 1 for enterprises in the above 17 heavy pollution industries and 0 for other enterprises.

3.2.3 Mediating variable

According to the previous analysis, there are two mediating variables in this paper, namely, technological innovation (*Re*) and agency costs (*Ag*). *Re* is replaced by the natural logarithm of Research and Development investment; *Ag* can be replaced by the total asset turnover rate (in reverse, the higher the total asset turnover rate, the lower the agency costs of the enterprise).

3.2.4 Control variable

Control variables are all kinds of main variables that affect the financial performance of enterprises, such as enterprise size (*Size*), enterprise age (*Age*), enterprise growth (*Gro*), operating cash flow ratio (*Ca*), number of employees (*Em*), equity multiplier (*Eq*), asset liability ratio (*Le*), chairman and general manager two in one (*One*), the sum of top three executives' salaries (*Top3_s*), equity concentration (*Top3*), and independent director ratio (*Ind_r*). In addition, the fixed effects of year (*Year*) and company (*Com*) are also added.

3.3 Big Data Model Setting

We built baseline regression Models 1–4 by combining difference-in-differences big data analysis technology.

$$ROA_{i,t} = \partial_0 + \partial_1 a_b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \xi_{i,t} \quad (1)$$

$$ROA_{i,t} = \partial_0 + \partial_1 a_b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \partial_4 C_{i,t} + \xi_{i,t} \quad (2)$$

$$ROA_{i,t} = \partial_0 + \partial_1 a_b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \partial_4 C_{i,t} + \partial_5 Year_t + \xi_{i,t} \quad (3)$$

$$ROA_{i,t} = \partial_0 + \partial_1 a_b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \partial_4 C_{i,t} + \partial_5 Year_t + \partial_6 Com_i + \xi_{i,t} \quad (4)$$

Model 1 is a regression model without any control variables, where ∂_0 is the intercept, and ∂_1 – ∂_6 are the regression coefficients to be estimated; $ROA_{i,t}$ is the dependent variable of this paper, enterprise financial performance; $Dif_{i,t}$ is the independent variable of this paper, that is, the difference-in-differences variable; and $\xi_{i,t}$ represents the regression residual. Model 2 adds a series of control variables $C_{i,t}$ that affect the financial performance of enterprises. Model 3 further increases the annual fixed effect $Year_t$. Model 4 finally adds the fixed effect of enterprises Com_i .

In order to test Hypothesis 2, that is, to test the mediating effect of technological innovation between the

implementation of environmental protection tax policies and corporate financial performance, we built models 5–7 based on Model 4 and combined with big data mechanism analysis technology.

$$Re_{i,t} = \partial_0 + \partial_1 a_{-}b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \partial_4 C_{i,t} + \partial_5 Year_t + \partial_6 Com_i + \xi_{i,t} \quad (5)$$

$$ROA_{i,t} = \partial_0 + \partial_1 Re_{i,t} + \partial_3 C_{i,t} + \partial_4 Year_t + \partial_5 Com_i + \xi_{i,t} \quad (6)$$

$$ROA_{i,t} = \partial_0 + \partial_1 a_{-}b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \partial_4 Re_{i,t} + \partial_5 C_{i,t} + \partial_6 Year_t + \partial_7 Com_i + \xi_{i,t} \quad (7)$$

In Models 5–7, $Re_{i,t}$ represents technological innovation, which serves as the mediating variable. If the coefficient of $Dif_{i,t}$ is significant in Models 4 and 5, the coefficient of $Re_{i,t}$ is significant in Model 6, and both $Re_{i,t}$ and $Dif_{i,t}$ are significant in Model 7, while the absolute value of the coefficient of $Dif_{i,t}$ in Model 7 is smaller than that in Model 4, technological innovation can be considered to have a partial mediating effect. These results indicate that the implementation of the environmental protection tax law improves enterprises' financial performance partly by promoting technological innovation.

In order to test Hypothesis 3, that is, to test the mediating effect of agency costs between the implementation of environmental protection tax policies and corporate financial performance, we built Models 8–10 based on Model 4 and combined with big data mechanism analysis technology.

$$Ag_{i,t} = \partial_0 + \partial_1 a_{-}b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \partial_4 C_{i,t} + \partial_5 Year_t + \partial_6 Com_i + \xi_{i,t} \quad (8)$$

$$ROA_{i,t} = \partial_0 + \partial_1 Ag_{i,t} + \partial_3 C_{i,t} + \partial_4 Year_t + \partial_5 Com_i + \xi_{i,t} \quad (9)$$

$$ROA_{i,t} = \partial_0 + \partial_1 a_{-}b_{i,t} + \partial_2 Im_{i,t} + \partial_3 Dif_{i,t} + \partial_4 Ag_{i,t} + \partial_5 C_{i,t} + \partial_6 Year_t + \partial_7 Com_i + \xi_{i,t} \quad (10)$$

In Models 8–10, $Ag_{i,t}$ is the mediating variable of agency costs. The test method of mediating effect of $Ag_{i,t}$ is similar to that of $Re_{i,t}$ mentioned above, and will not be repeated here.

4. Empirical Analysis of Big Data

4.1 Descriptive Statistical Analysis of Big Data

Table 1. Descriptive statistics of main variables

Variable	Mean	SD	Min.	P50	Max.
<i>ROA</i>	0.049	0.073	−0.227	0.045	0.280
<i>ROE</i>	0.073	0.157	−0.793	0.080	0.475
<i>Re</i>	17.861	1.538	7.278	17.793	25.025
<i>Ag</i>	0.645	0.428	0.073	0.554	2.622
<i>Size</i>	22.112	1.422	14.942	21.951	28.791
<i>Age</i>	2.923	0.354	0.693	2.944	4.248
<i>Gro</i>	0.142	0.406	−0.589	0.082	2.708
<i>Ca</i>	0.049	0.070	−0.172	0.048	0.258
<i>Em</i>	7.589	1.299	0.000	7.502	13.784
<i>Eq</i>	2.069	1.331	1.054	1.661	9.874
<i>Le</i>	0.410	0.205	0.052	0.398	0.899
<i>One</i>	0.319	0.466	0.000	0.000	1.000
<i>Top3_s</i>	14.691	0.714	9.385	14.661	18.584
<i>Top3</i>	0.490	0.157	0.006	0.486	0.980
<i>Ind_r</i>	0.378	0.055	0.143	0.364	0.818

Note: SD = standard deviation; P50 = 50th percentile (median); *ROA* = total net asset interest rate; *ROE* = return on equity; *Re* = technological innovation; *Ag* = agency costs; *Size* = enterprise size; *Age* = enterprise age; *Gro* = enterprise growth; *Ca* = operating cash flow ratio; *Em* = number of employees; *Eq* = equity multiplier; *Le* = asset liability ratio; *One* = chairman and general manager two in one; *Top3_s* = the sum of top three executives' salaries; *Top3* = equity concentration; *Ind_r* = independent director ratio.

In this paper, the big data management technology is applied to list the distribution of all variables in Table 1. From Table 1, it can be seen that the data selected by the big data filtering technology in this paper is relatively

reliable and has no obvious outliers.

4.2 Big Data Baseline Regression Analysis

The baseline regression results processed by the difference-in-differences big data analysis technology in this paper are shown in Table 2.

According to the research method of difference-in-differences big data analysis technology, column Model 1 only tests the influence of independent variables on dependent variables, without any control variables, and its *Dif* coefficient is 0.017; A series of control variables affecting the dependent variable are added in column Model 2, and the *Dif* coefficient is 0.009; Model 3 column has added annual fixed effect, and its *Dif* coefficient is 0.009; Model 4 column further increases the fixed effect of enterprises, and its *Dif* coefficient is 0.008. The *Dif* coefficients of Model 1–4 columns are significant at the level of 1%. This shows that there is a significant positive correlation between the implementation of environmental protection tax policies and corporate financial performance, and Hypothesis 1 passed the test. Since the variables in Model 4 column are more complete, the subsequent research will be carried out on the basis of Model 4 column.

Table 2. The results of baseline regression analysis

Variables	Model 1	Model 2	Model 3	Model 4
<i>Dif</i>	0.017*** (15.51)	0.009*** (8.29)	0.009*** (8.19)	0.008*** (7.65)
Control variables	No	Yes	Yes	Yes
Year	No	No	Yes	Yes
Com	No	No	No	Yes

Note: *Dif*= the difference-in-differences interaction term measuring the effect of the environmental protection tax policy; Year = fixed effects of year; Com = fixed effects of company; * indicates that the significant is at the 10% level, ** indicates that the significant is at the 5% level, *** indicates that the significant is at the 1% level, *t* value of variables is in the brackets.

4.3 Big Data Robustness Test

4.3.1 Big data time placebo test

Referring to the analysis method of Wang (2024), we apply big data generation technology to assume that the implementation time of the environmental protection tax law policy is 1 year ahead of schedule, 2 years ahead of schedule, 1 year behind schedule, and 2 years behind schedule, and then generate a new policy implementation time variable *a_b_n* and a new interaction term *Dif_n* for regression estimation, the results are shown in Table 3.

Table 3. The results of time placebo test

Variables	1	2	3	4
<i>Dif_n</i>	0.017 (1.18)	0.021 (1.35)	0.012 (0.88)	0.014 (1.07)
Control variables	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Com	Yes	Yes	Yes	Yes

Note: *Dif*= the difference-in-differences interaction term measuring the effect of the environmental protection tax policy; Year = fixed effects of year; Com = fixed effects of company.

The regression results in Table 3 tell us that after applying big data generation technology to change the implementation time of the policy, the coefficients of *Dif_n* variables become insignificant, which proves that our previous regression analysis results are robust.

4.3.2 Big data individual placebo test

In order to test whether the baseline regression results will be affected by random interference, this paper constructs a two-level random experiment of virtual policy affected enterprises and virtual policy implementation time by randomly selecting the implementation group of enterprises and randomly changing the policy implementation time based on big data random screening technology. According to the multiplication term *Dif_sub* of the two to replace the original interaction term for estimation, in order to enhance the reliability of the experimental results, we will conduct 500 random experiments. Finally, we use data visualization technology to show the distribution of *Dif_sub* variable coefficients in Figure 2.

As shown in Figure 2, *Dif_sub* variable coefficients presented by data visualization technology are concentrated around 0, and their distribution conforms to normal distribution. The significance level of coefficient regression results is mostly above 0.1, that is, it is not significant at the level of 10%. The *Dif* variable coefficient 0.008 in

the baseline regression result Model 4 is at the high tail of the coefficient distribution in Figure 2, which is defined as a small probability event in Econometrics and Statistics. This shows that our baseline regression results are not affected by random interference, and the research results are robust.

4.3.3 Excluding samples from 2020 to 2022

Since the sample period (2012–2024) of this paper covers a number of major exogenous shocks, especially the new crown epidemic (2020–2022), which has seriously disrupted the supply chain, market demand and government law enforcement, it may confuse the estimated results of the effect of environmental protection tax. Therefore, we removed the samples from 2020 to 2022 to analyze Models 1–4, and the analysis results are shown in Table 4.

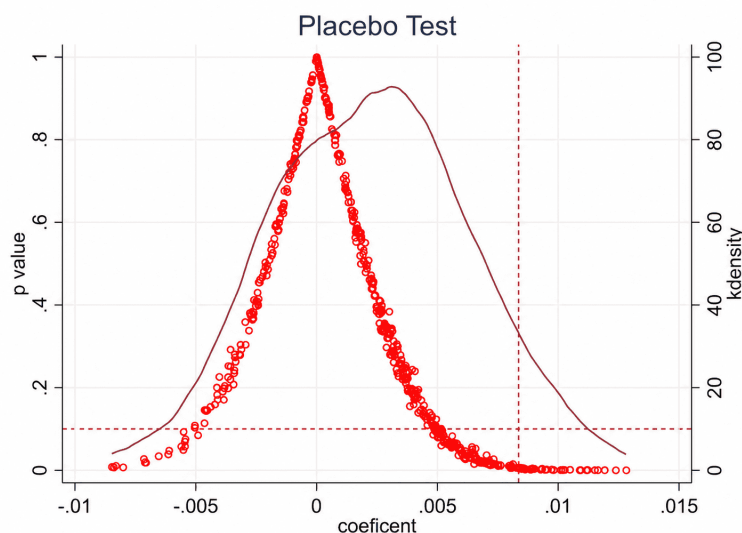


Figure 2. The results of placebo test

Table 4. Regression results excluding 2020–2022

Variables	Model 1	Model 2	Model 3	Model 4
<i>Dif</i>	0.014*** (11.38)	0.009*** (7.38)	0.008*** (7.14)	0.008*** (6.45)
Control variables	No	Yes	Yes	Yes
Year	No	No	Yes	Yes
Com	No	No	No	Yes

Note: *Dif* = the difference-in-differences interaction term measuring the effect of the environmental protection tax policy; Year = fixed effects of year; Com = fixed effects of company; * indicates that the significant is at the 10% level, ** indicates that the significant is at the 5% level, *** indicates that the significant is at the 1% level, *t* value of variables is in the brackets.

From the results in Table 4, it can be seen that the *Dif* coefficients are all positive and have passed the significance level test of 1%, which is consistent with the results shown in Table 1. It can be seen that the regression results of this paper are still robust after excluding the data during the new crown epidemic.

4.4 Analysis of Mediation Effect of Big Data

4.4.1 Analysis of mediating effect of technological innovation

In order to test the mediating effect of technological innovation, we applied the big data mechanism analysis technology to make regression estimation for Models 4–7, and the results are shown in Table 5.

In Table 5 presented by big data mechanism analysis technology, the *Dif* coefficient of Model 4 column is 0.008, which is significant at the level of 1%, indicating that the implementation of environmental protection tax law policy is positively correlated with the financial performance of enterprises; The coefficient of *Dif* in Model 5 column is 0.083 and is significant at the level of 1%, indicating that there is a positive correlation between the implementation of environmental protection tax law and enterprise technological innovation; The *Re* variable coefficient of Model 6 column is 0.002 and is significant at the level of 1%, indicating that there is a positive correlation between technological innovation and enterprise financial performance; The *Re* variable and *Dif* variable coefficients of Model 7 column are positive and both pass the significance level test of 1%, and the *Dif*

coefficient is 0.006, and its absolute value is less than the absolute value of the *Dif* coefficient of Model 4 column 0.008, which shows that the implementation of the environmental protection tax law policy can promote enterprises to improve the level of technological innovation and improve the financial performance of enterprises. Hypothesis 2 is verified.

Table 5. Mediating effect of technological innovation

Variables	Model 4	Model 5	Model 6	Model 7
<i>Dif</i>	0.008*** (7.65)	0.083*** (6.04)	—	0.006*** (5.22)
<i>Re</i>	—	—	0.002*** (3.47)	0.002*** (3.29)
Control variables	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Com	Yes	Yes	Yes	Yes

Note: *Dif* = the difference-in-differences interaction term measuring the effect of the environmental protection tax policy; *Re* = technological innovation; Year = fixed effects of year; Com = fixed effects of company; * indicates that the significant is at the 10% level, ** indicates that the significant is at the 5% level, *** indicates that the significant is at the 1% level, *t* value of variables is in the brackets; an en-dash (—) indicates not applicable.

4.4.2 The mediating effect analysis of agency costs

To test the mediating effect of agency costss, we used the big data mechanism analysis technology to make regression estimates for Model 4 and Models 8–10, and the results are shown in Table 6.

In Table 6 presented by big data mechanism analysis technology, column Model 4 is the same as column Model 4 in Table 5; The *Dif* coefficient of Model 8 column is 0.023, and it is significant at the level of 1%, which indicates that the implementation of environmental protection tax law policy is negatively correlated with the agency costs of enterprises (because the total asset turnover rate is a negative alternative indicator of agency costs); The *Ag* coefficient of Model 9 column is 0.031 and highly significant, which indicates that the agency costs is negatively correlated with the financial performance of enterprises; Both *Ag* and *Dif* coefficients in column Model 10 are positive and highly significant. The coefficient of *Dif* is 0.0077, and its absolute value is less than the absolute value of *Dif* coefficient 0.0084 in column Model 4, which indicates that the implementation of environmental protection tax law policy can reduce the agency costs of enterprises and have a positive impact on the financial performance of enterprises. Hypothesis 3 is proved.

Table 6. Mediating effect of agency costs

Variables	Model 4	Model 8	Model 9	Model 10
<i>Dif</i>	0.0084*** (7.65)	0.023*** (5.03)	—	0.0077*** (7.06)
<i>Ag</i>	—	—	0.031*** (24.56)	0.031*** (24.39)
Control variables	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Com	Yes	Yes	Yes	Yes

Note: *Dif* = the difference-in-differences interaction term measuring the effect of the environmental protection tax policy; *Ag* = agency costs; Year = fixed effects of year; Com = fixed effects of company; * indicates that the significant is at the 10% level, ** indicates that the significant is at the 5% level, *** indicates that the significant is at the 1% level, *t* value of variables is in the brackets; an en-dash (—) indicates not applicable; the *Dif* coefficients of Model 4 and Model 10 columns should be kept to four digits after the decimal point for comparison.

4.4.3 Endogenous test of mediating effect

We use standard regression analysis to test the mediating effect, but such analysis may face the problems of reverse causality and missing variables. In order to alleviate the endogenous problem, we use the instrumental variable method to analyze. Referring to the research methods of Wu & Li (2025) and Zhao et al. (2024), we selected the average level of technological innovation and agency costss in the same year and industry (excluding the enterprise itself), as the tool variable, expressed by the instrumental variable for *Re* (IVRE) and the instrumental variable for *Ag* (IVAG) respectively, and applied the two-stage least square method (2SLS) to carry out the endogenous test. The test results are listed in Table 7.

As shown in Table 7, the coefficients of IVRE and IVAG in the first column are significant at the 1% level, and the *Re* and *Ag* variables in the second column are consistent with the coefficients in the Model 6 column of Table 5 and the Model 9 column of Table 6, and both are significant at the 1% level. Anderson canon. corr. LM statistic is significant, and Cragg Donald Wald F statistic is also far greater than the critical value of 10%. This means that after alleviating the endogenous, the mediating effect of technological innovation and agency costss still exists.

Table 7. Endogenous test of mediating variables

Variables	Technological Innovation		Agency Costs	
	First	Second	First	Second
IVRE	0.227*** (0.011)	—	—	—
<i>Re</i>	—	0.013*** (0.004)	—	—
IVAG	—	—	0.425*** (0.011)	—
<i>Ag</i>	—	—	—	0.044*** (0.006)
Control variables	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Com	Yes	Yes	Yes	Yes
Anderson canonical correlation LM statistic	434.728***		1558.323***	
Cragg-Donald Wald <i>F</i> statistic	440.532 [16.38]		1628.251 [16.38]	

Note: IVRE = the instrumental variable for *Re*; *Re* = technological innovation; IVAG = the instrumental variable for *Ag*; *Ag* = agency costs; Year = fixed effects of year; Com = fixed effects of company; LM = Lagrange multiplier; * indicates that the significant is at the 10% level, ** indicates that the significant is at the 5% level, *** indicates that the significant is at the 1% level, *t* values of variables are in the brackets; an en-dash (—) indicates not applicable.

5. Conclusions

Whether the implementation of the environmental protection tax law policy can promote the economic benefits of enterprises has been a concern of the academic community. This paper applies difference-in-differences big data analysis technology and big data mechanism analysis technology empirically studies the relationship between the implementation of environmental protection tax law and corporate financial performance. The research conclusions are as follows: there is a significant positive correlation between the implementation of environmental protection tax law and corporate financial performance; The implementation of environmental protection tax policy can promote enterprises to improve the level of technological innovation and reduce the agency costs of enterprises, which has a positive effect on the financial performance of enterprises.

According to the research conclusion and big data analysis results of this paper, we put forward the following policy suggestions:

(1) In order to ensure the effective implementation of the environmental protection tax law, it is suggested to build a limited data sharing mechanism between tax and environmental protection departments on the basis of the existing environmental credit evaluation system. At the initial stage, we can focus on the cross checking of the list of key monitored enterprises, the implementation data of emission permits and the tax collection and management records, and use the rule engine instead of complex algorithms to carry out risk prompts (such as the abnormal comparison of emission data and environmental protection tax declaration data). At the same time, the authority and confidentiality obligations are clarified by signing a departmental data sharing agreement to avoid premature involvement in complex projects such as cross departmental federal modeling or algorithm fairness, and ensure that the scheme is promoted step by step within the scope of legal authorization and technical maturity.

(2) In order to improve the guiding effectiveness of tax incentives, we can build a dynamic and multi-level tax preference model based on big data analysis. According to the real-time emission data of enterprises, the computer algorithm automatically matches the progressive reduction grade of “the less emission, the more preferential”, and accurately quantifies the space for enterprises’ environmental protection investment and cost reduction and benefit increase. This mechanism can not only use data insight to stimulate the endogenous power of green innovation of enterprises, but also reduce the agency costs of environmental compliance through automated supervision, and ultimately drive the steady improvement of enterprise financial performance.

Author Contributions

Conceptualization, J.W. and J.L.; methodology, J.W.; software, J.W.; validation, Y.L., Y.W., and J.J.L.; formal analysis, J.W.; investigation, J.J.L.; resources, Y.L.; data curation, J.W.; writing—original draft preparation, J.W.; writing—review and editing, Y.L.; visualization, J.L.; supervision, Y.W.; project administration, J.W. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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